

FUNDAMENTALS OF WIRELESS COMMUNICATION TECHNOLOGY

- **THE ELECTROMAGNETIC SPECTRUM**

Waves can be characterized by their frequency (f) and wavelength (λ). Heinrich Hertz, the German physicist who discovered radio, and wavelength is the distance between two consecutive maxima or minima in the wave.

The speed of propagation of these waves (c) varies from medium to medium.

Except in a vacuum where all electromagnetic waves travel at the same speed, the speed of light.

$$c = \lambda * f$$

frequency bands in the electromagnetic spectrum as defined by the International Telecommunications Union (ITU). ITU, located in Geneva and a suborganization of the United Nations, coordinates wired and wireless telecommunication activities worldwide.

Frequency bands and their common uses

Band Name	Frequency	Wavelength	Applications
Extremely Low Frequency (ELF)	30 to 300 Hz	10,000 to 1,000 Km	Powerline frequencies
Voice Frequency (VF)	300 to 3,000 Hz	1,000 to 100 Km	Telephone communications
Very Low Frequency (VLF)	3 to 30 KHz	100 to 10 Km	Marine communications
Low Frequency (LF)	30 to 300 KHz	10 to 1 Km	Marine communications
Medium Frequency (MF)	300 to 3,000 KHz	1,000 to 100 m*	AM broadcasting
High Frequency (HF)	3 to 30 MHz	100 to 10 m	Long-distance aircraft/ship communications
Very High Frequency (VHF)	30 to 300 MHz	10 to 1 m	FM broadcasting
Ultra High Frequency (UHF)	300 to 3,000 MHz	100 to 10 cm	Cellular telephone
Super High Frequency (SHF)	3 to 30 GHz	10 to 1 cm	Satellite communications, microwave links
Extremely High Frequency (EHF)	30 to 300 GHz	10 to 1 mm	Wireless local loop
Infrared	300 GHz to 400 THz	1 mm to 770 nm	Consumer electronics
Visible Light	400 THz to 900 THz	770 nm to 330 nm	Optical communications

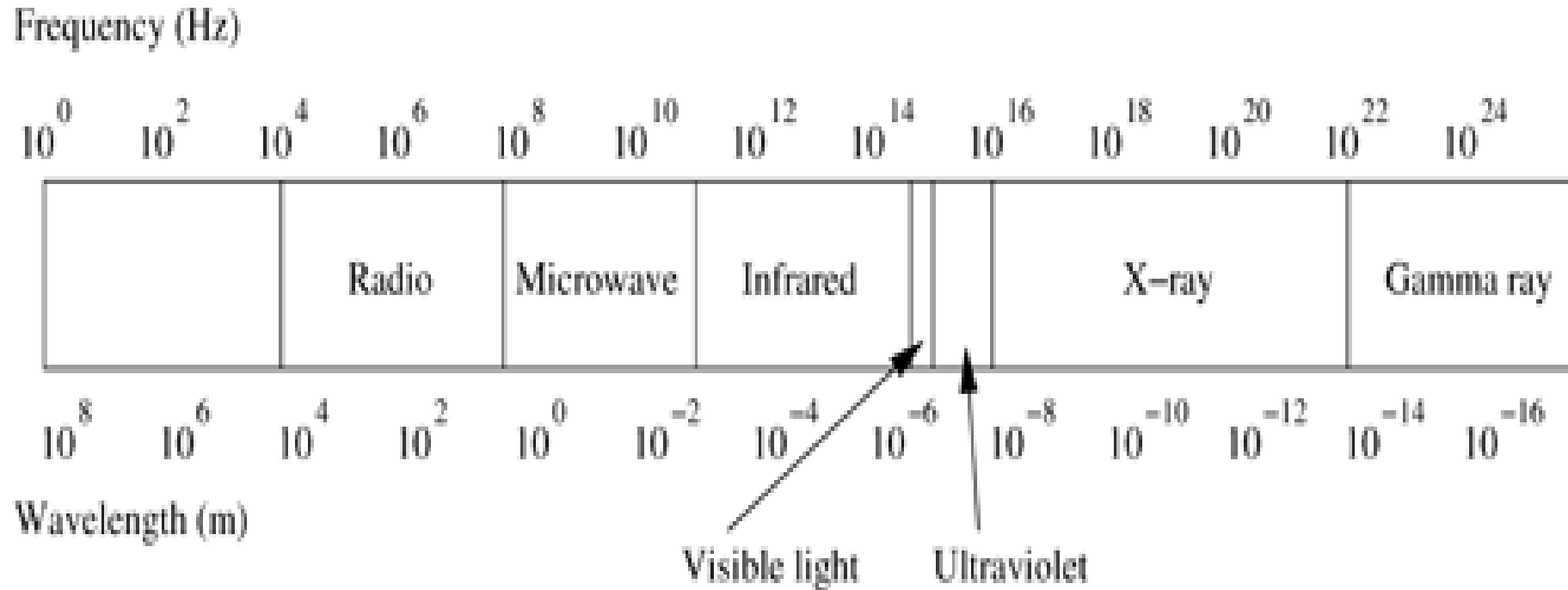
* Throughout this book, the unit *m* refers to meter(s).

Speed, Wavelength, Frequency

Light speed = Wavelength x Frequency

$$= 3 \times 10^8 \text{ m/s} = 300,000 \text{ km/s}$$

System	Frequency	Wavelength
AC current	60 Hz	5,000 km
FM radio	100 MHz	3 m
Cellular	800 MHz	37.5 cm
Ka band satellite	20 GHz	15 mm
Ultraviolet light	10^{15} Hz	10^{-7} m



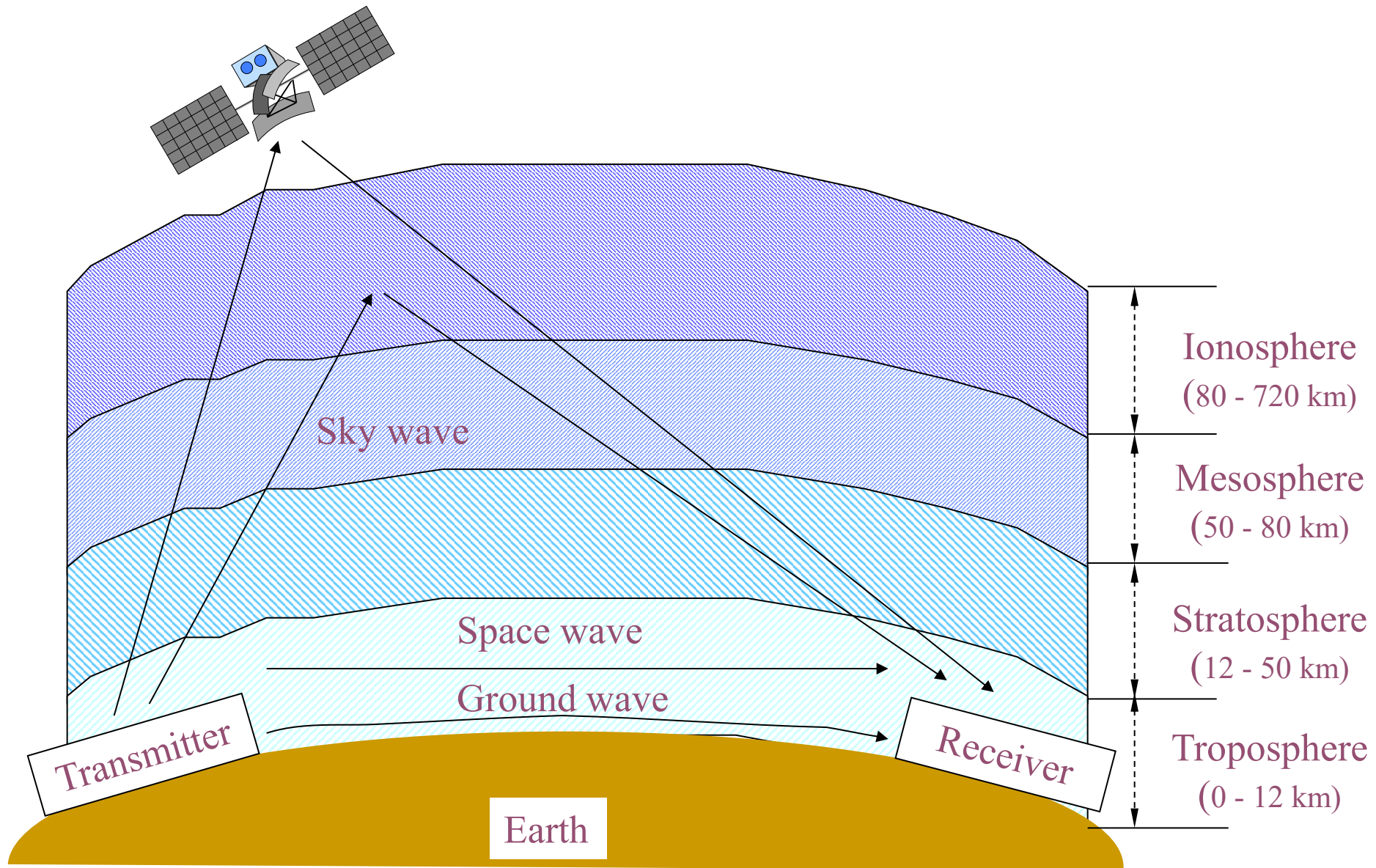
Low-frequency bands comprised of the radio, microwave, infrared, and visible light portions of the spectrum can be used for information transmission by modulating the amplitude, frequency, or the phase of the waves (the different modulation techniques)

The high frequency waves such as X-rays and Gamma rays, theoretically better for information propagation, are not used due to practical concerns such as the difficulty to generate and modulate these waves, and the harm they could cause to living things. Also, such high-frequency waves do not propagate well through buildings.

- Radio waves are easy to generate and used for both indoor and outdoor communication.
- They have ability to pass through buildings and ability to travel long distances.
- Radio transmission is omnidirectional (when radio waves are generated, they spread out from the transmitting antenna in all directions) in nature, the need to physically align the transmitter and receiver also does not arise.
- The frequency of the radio wave determines many of the characteristics of the transmission.
- At low frequencies the waves can pass through obstacles easily, but their power falls with an inverse-squared relation with respect to the distance.
- The higher frequency waves are more prone to absorption by rain drops, and they get reflected by obstacles.
- Due to the long transmission range of the radio waves, interference between transmissions is a problem that needs to be addressed.

- VLF, LF, and MF bands the propagation of waves, also called as ground waves, follows the curvature of the Earth.
- The maximum transmission ranges of these waves are of the order of a few hundred kilometers.
- They are used for low bandwidth transmissions such as amplitude modulated (AM) radio broadcasting.
- The HF and VHF band transmissions are absorbed by the atmosphere near the Earth's surface. However, a portion of the radiation, called the sky wave, radiates outward and upward to the ionosphere in the upper atmosphere.
- The ionosphere contains ionized particles formed due to the sun's radiation. These ionized particles reflect the sky waves back to the Earth. A powerful sky wave may get reflected several times between the Earth and the ionosphere.
- Sky waves are used by amateur ham radio operators and for military communication.

Types of Waves



Radio Frequency Bands

Classification Band	Initials	Frequency Range	Characteristics
Extremely low	ELF	< 300 Hz	Ground wave
Infra low	ILF	300 Hz - 3 kHz	
Very low	VLF	3 kHz - 30 kHz	
Low	LF	30 kHz - 300 kHz	
Medium	MF	300 kHz - 3 MHz	Ground/Sky wave
High	HF	3 MHz - 30 MHz	Sky wave
Very high	VHF	30 MHz - 300 MHz	Space wave
Ultra high	UHF	300 MHz - 3 GHz	
Super high	SHF	3 GHz - 30 GHz	
Extremely high	EHF	30 GHz - 300 GHz	
Tremendously high	THF	300 GHz - 3000 GHz	

- Microwave transmissions (in the SHF band) tend to travel in straight lines and hence can be narrowly focused.
- Microwaves were widely used for long-distance telephony, before they got replaced by fiber optics.
- They are also widely used for mobile phones and television transmission. Since the energy is concentrated into a small beam, these transmissions have a higher signal-to-noise ratio (SNR).
- SNR is the ratio of the signal power to the noise power on a transmission medium, and is used to categorize the quality of a transmission.
- Higher frequency of operation they do not pass through buildings. Hence, proper line-of-sight alignment between the transmitting and the receiving antennas is required.
- Straight-line microwave transmission over long distances requires repeaters at regular distances as microwaves get attenuated by objects found in their path.
- The number of repeaters required is a function of the microwave transmission tower height.

- Infrared waves and waves in the EHF band (also known as millimeter waves) are used for short-range communication.
- They are widely used in television, VCR, and stereo remote controls.
- They are relatively directional and inexpensive to build.
- They cannot travel through obstacles, which is a light-like (and not radio-like) property that is observed in very high-frequency transmissions.
- The visible light part of the spectrum is just after the infrared portion. Unguided optical signaling using visible light provides very high bandwidth at a very low cost.
- Recent applications of light-wave transmissions involve the use of lasers to connect LANs on two buildings through roof-top antennas.
- But the main disadvantage here is that it is very difficult to focus a very narrow unidirectional laser beam, which limits the maximum distance between the transmitter and receiver. Also, such waves cannot penetrate through rain or thick fog.

Spectrum Allocation

Electromagnetic spectrum is a common resource which is open for access by anyone, National and international agreements have been drawn regarding the usage of the different frequency bands within the spectrum.

The individual national governments allocate spectrum for applications such as AM/FM radio broadcasting, television broadcasting, mobile telephony, military communication, and government usage.

Worldwide, an agency of the International Telecommunications Union Radiocommunication (ITU-R) Bureau called World Administrative Radio Conference (WARC) tries to coordinate the spectrum allocation by the various national governments, so that communication devices that can work in multiple countries can be manufactured.

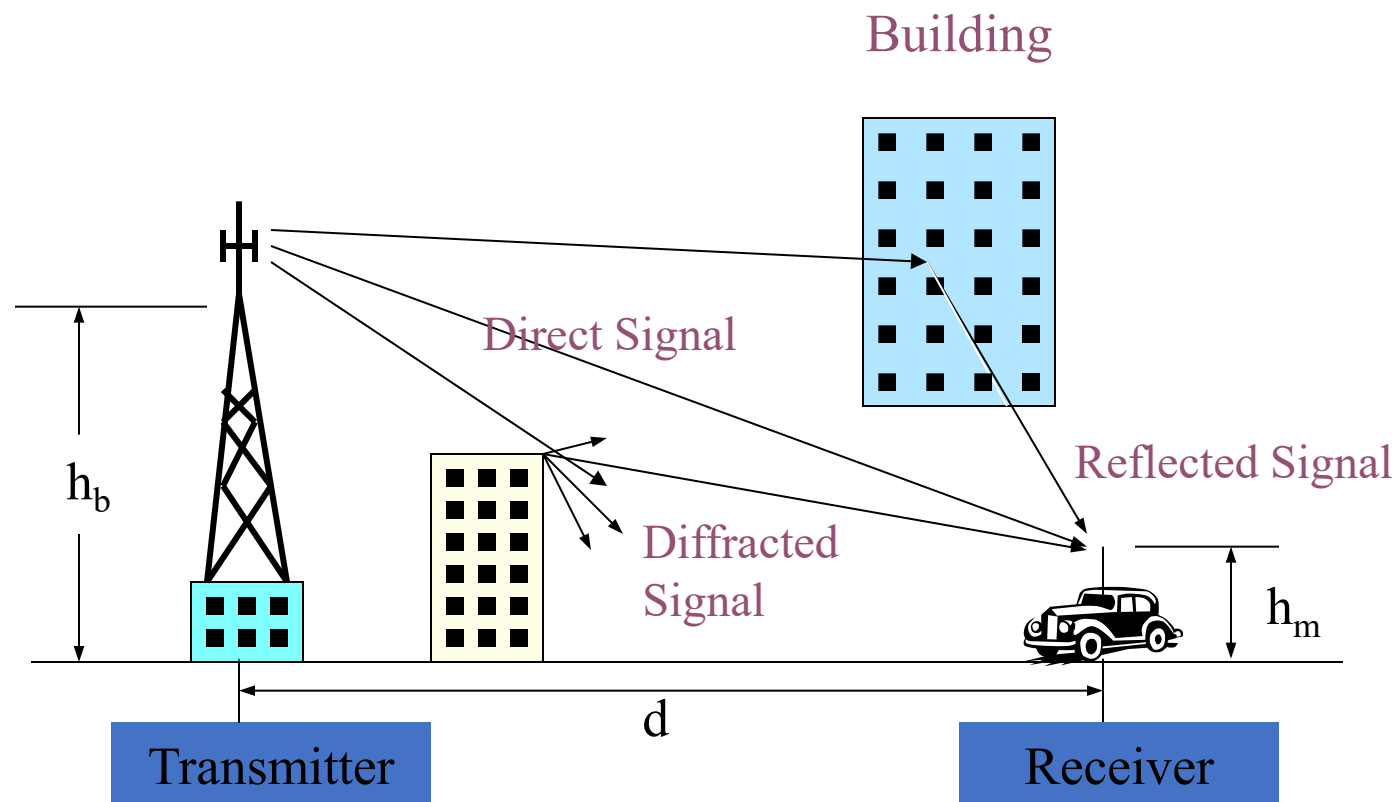
However, the recommendations of ITU-R are not binding on any government. Regarding the national spectrum allocation, even when the government sets aside a portion of the spectrum for a particular application/use (*e.g.*, cellular telephony)

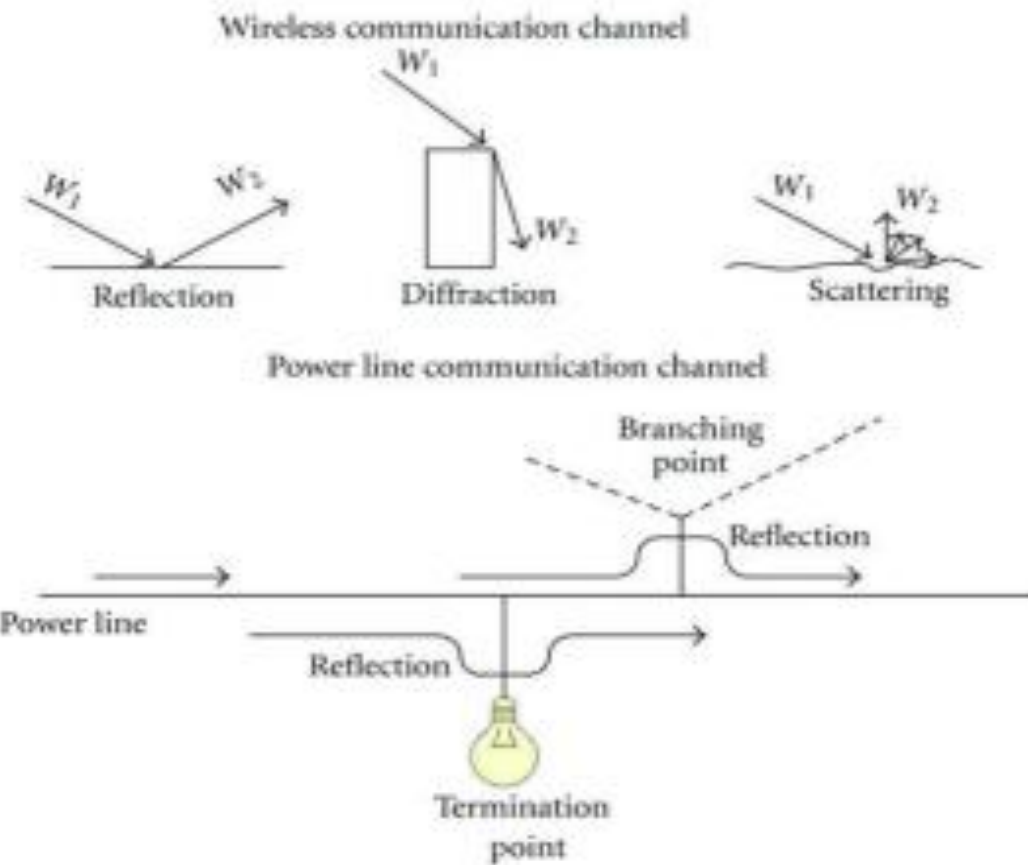
Another issue crops up – the issue of which carrier (company) is to use which set of frequencies. Many methods have been tried out for this frequency allocation among multiple competing carriers, some of which are described below.

Radio Propagation Mechanisms

- Reflection
 - Propagation wave impinges on an object which is large as compared to wavelength
 - e.g., the surface of the Earth, buildings, walls, etc.
- Diffraction
 - Radio path between transmitter and receiver obstructed by surface with sharp irregular edges
 - Waves bend around the obstacle, even when LOS (line of sight) does not exist
- Scattering
 - Objects smaller than the wavelength of the propagation wave
 - e.g. foliage, street signs, lamp posts

Radio Propagation Mechanism





Propagation mechanisms for wireless and PLC channels.

CHARACTERISTICS OF THE WIRELESS CHANNEL

The wireless channel (transmission medium) is susceptible to a variety of transmission impediments such as path loss, interference, and blockage. These factors restrict the range, data rate, and the reliability of the wireless transmission.

These factors affect the transmission depends upon the environmental conditions and the mobility of the transmitter and receiver.

Transmitted signal has a direct-path component between the transmitter and the receiver. Other components of the transmitted signal known as multipath components are reflected, diffracted, and scattered by the environment, and arrive at the receiver shifted in amplitude, frequency, and phase with respect to the direct path component

Characteristics of the wireless channel are path loss, fading, interference, and Doppler shift.

Two key constraints, Nyquist's and Shannon's theorems, that govern the ability to transmit information at different data rates

Path Loss

free space model: path loss can be expressed ratio of the power of the transmitted signal to the power of the same signal received by the receiver, on a given path. It is a function of the propagation distance.

Estimation of path loss is very important for designing and deploying wireless communication networks.

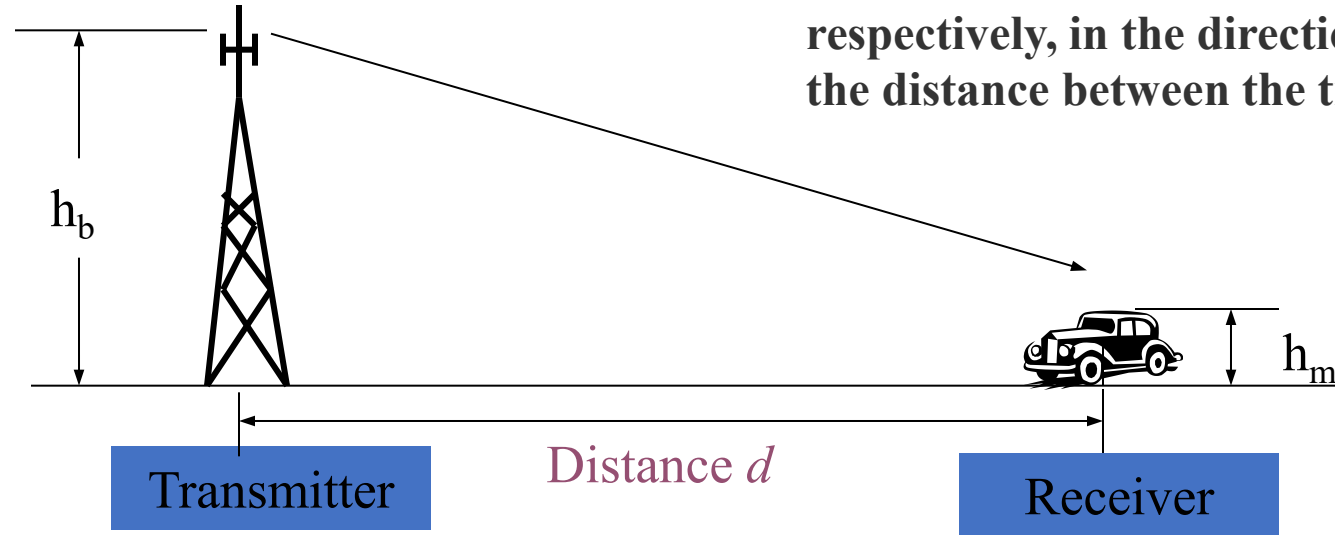
Path loss is dependent on a number of factors such as the radio frequency used and the nature of the terrain.

This model can not determine the characteristics of every transmission.

Therefore, in designing a network, several models are required to describe the variety of transmission environments.

The *free space* propagation model is the simplest path loss model in which there is a direct-path signal between the transmitter and the receiver, with no atmospheric attenuation or multipath components.

Free-space Propagation



where G_t and G_r are the transmitter and receiver antenna gains, respectively, in the direction from the transmitter to the receiver, d is the distance between the transmitter and receiver, and $\lambda = c/f$

- The received signal power at distance d :

$$P_r = \frac{A_e G_t P_t}{4\pi d^2}$$

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi d} \right)^2$$

where P_t is transmitting power, A_e is effective area, and G_t is the transmitting antenna gain. Assuming that the radiated power is uniformly distributed over the surface of the sphere.

Antenna Gain

Antenna gain is defined as the ratio of the power required at the input of a loss-free reference antenna to the power supplied to the input of the given antenna to produce, in a given direction, signals of the same strength at the same distance

- For a circular reflector antenna

$$\text{Gain } G = \eta (\pi D / \lambda)^2$$

η = net efficiency (depends on the electric field distribution over the antenna aperture, losses, ohmic heating, typically 0.55)

D = diameter

thus, $G = \eta (\pi D f / c)^2$, $c = \lambda f$ (c is speed of light)

Example:

- Antenna with diameter = 2 m, frequency = 6 GHz, wavelength = 0.05 m

$$G = 39.4 \text{ dB}$$

- Frequency = 14 GHz, same diameter, wavelength = 0.021 m

$$G = 46.9 \text{ dB}$$

* Higher the frequency, higher the gain for the same size antenna

Two-ray path loss model

This popular path loss model is also called the *two-path* model.

The free space model described above assumes that there is only one single path from the transmitter to the receiver.

But in reality, the signal reaches the receiver through multiple paths (because of reflection, refraction, and scattering). The two-path model tries to capture this phenomenon.

The model assumes that the signal reaches the receiver through two paths, one a line-of sight path, and the other the path through which the reflected (or refracted, or scattered) wave is received.

According to the two-path model, the received power is given by

$$P_r = P_t G_t G_r \left(\frac{h_t h_r}{d^2} \right)^2$$

where P_t is the transmitted power, G_t and G_r represent the antenna gains at the transmitter and the receiver, respectively, d is the distance between the transmitter and receiver, and h_t and h_r are the heights of the transmitter and the receiver, respectively.

In the general case, for isotropic antennas (antennas in which the power of the transmitted signal is the same in all directions), the received power of the signal is given by

$$P_r = P_t G_t G_r \left(\frac{\lambda}{4\pi} \right)^2 \frac{1}{d^\gamma}$$

where γ is the propagation coefficient that varies between 2 (free-space propagation) and 5 (strong attenuation).

Fading

Fading refers to the fluctuations in signal strength when received at the receiver. Fading can be classified into two types:

*fast fading/small-scale fading, and
Slow fading/large-scale fading*

Fast fading refers to the rapid fluctuations in the amplitude, phase, or multipath delays of the received signal, due to the interference between multiple versions (copies) of the same transmitted signal arriving at the receiver at slightly different times.

The time between the reception of the first version of the signal and the last echoed signal is called *delay spread*. The multipath propagation of the transmitted signal, which causes fast fading, is because of the three propagation mechanisms described previously, namely, **reflection, diffraction, and scattering**.

The multiple signal paths may sometimes add constructively or sometimes destructively at the receiver, causing a variation in the power level of the received signal.

The received signal envelope of a fast-fading signal is said to follow a Rayleigh distribution , if there is no line-of-sight path between the transmitter and the receiver (applicable in outdoor environments), and a Ricean distribution if one such path is available (characterizes indoor settings).

Slow fading occurs when objects that partially absorb the transmissions lie between the transmitter and receiver. Slow fading is so called because the duration of the fade may last for multiple seconds or minutes.

Slow fading may occur when the receiver is inside a building and the radio wave must pass through the walls of a building, or when the receiver is temporarily shielded from the transmitter by a building.

The obstructing objects cause a random variation in the received signal power. Slow fading may cause the received signal power to vary, though the distance between the transmitter and receiver remains the same.

Slow fading is also referred to as *shadow fading* since the objects that cause the fade, which may be large buildings or other structures, block the direct transmission path from the transmitter to the receiver.

Interference:

Wireless transmissions have to counter interference from a wide variety of sources. Two main forms of interference *are adjacent channel interference and co-channel interference.*

In the **adjacent channel interference** case, signals in nearby frequencies have components outside their allocated ranges, and these components may interfere with on-going transmissions in the adjacent frequencies. It can be avoided by carefully introducing guard bands² between the allocated frequency ranges.

Co-channel interference, sometimes also referred to as narrow-band interference, is due to other nearby systems (say, AM/FM broadcast) using the same transmission frequency. Narrow-band interference due to frequency reuse in cellular systems can be minimized with the use of multiuser detection mechanisms, directional antennas, and dynamic channel allocation methods.

A **guard band** is a small frequency band used to separate two adjacent frequency bands in order to avoid interference between them.

Multiuser detection is an effective approach used to combat the multiuser interference problems inherent in CDMA systems

Inter-symbol interference is another type of interference, where distortion in the received signal is caused by the temporal spreading and the consequent overlapping of individual pulses in the signal.

This temporal spreading of individual pulses (delay spread) goes above a certain limit (symbol detection time), the receiver becomes unable to reliably distinguish between changes of state in the signal, that is, the bit pattern interpreted by the receiver is not the same as that sent by the sender.

Doppler Shift

The Doppler shift is defined as the change/shift in the frequency of the received signal when the transmitter and the receiver are mobile with respect to each other.

If they are moving toward each other, then the frequency of the received signal will be higher than that of the transmitted signal, and if they are moving away from each other, the frequency of the signal at the receiver will be lower than that at the transmitter.

$$f_d = \frac{v}{\lambda}$$

The Doppler shift f_d is given by

where v is the relative velocity between the transmitter and receiver, and λ is the wavelength of the signal

Transmission Rate Constraints

Two important constraints that determine the maximum rate of data transmission on a channel are **Nyquist's theorem and Shannon's theorem**

The *Nyquist theorem* gives the maximum data rate possible on a channel. If B is the bandwidth of the channel (in Hz) and L is the number of discrete signal levels/voltage values used, then the maximum channel capacity C according to the Nyquist theorem is given by

$$C = 2 \times B \times \log_2 L \quad \text{bits/sec} \quad \text{The above condition is valid only for a noiseless channel.}$$

The number of changes per second is measured in terms of *baud*.

Noise level in the channel is represented by the SNR. It is the ratio of signal power (S) to noise power (N), specified in decibels, that is, $SNR = 10 \log_{10}(S/N)$.

One of the most important contributions of Shannon was his theorem on the maximum data rate possible on a noisy channel. According to *Shannon's theorem*, the maximum data rate C is given by

$$C = B \times \log_2(1 + (S/N)) \quad \text{bits/sec}$$

where B is the bandwidth of the channel (in Hz).

WIRELESS LANS AND PANS

These privately-owned networks are known as local area networks (LANs) which come under the category of small-scale networks (networks within a single building or campus with a size of a few kilometers).

To do away with the wiring associated with the interconnection of PCs in LANs, researchers have explored the possible usage of radio waves and infrared light for interconnection.

This has resulted in the emergence of wireless LANs (WLANs), where wireless transmission is used at the physical layer of the network. Wireless personal area networks (WPANs) are the next step down from WLANs, covering smaller areas with low power transmission, for networking of portable and mobile computing devices such as PCs, personal digital assistants (PDAs).

FUNDAMENTALS OF WLANS

Technical Issues

Differences Between Wireless and Wired Transmission

- **Address is not equivalent to physical location:** In a wireless network, address refers to a particular station and this station need not be stationary. Therefore, address may not always refer to a particular geographical location.
- **Dynamic topology and restricted connectivity:** The mobile nodes may often go out of reach of each other. This means that network connectivity is partial at times.
- **Medium boundaries are not well-defined:** The exact reach of wireless signals cannot be determined accurately. It depends on various factors such as signal strength and noise levels. This means that the precise boundaries of the medium cannot be determined easily.
- **Error-prone medium:** Transmissions by a node in the wireless channel are affected by simultaneous transmissions by neighboring nodes that are located within the direct transmission range of the transmitting node. This means that the error rates are significantly higher in the wireless medium.

Goals which have to be achieved while designing WLANs

License-free operation: One of the major factors that affects the cost of wireless access is the license fee for the spectrum in which a particular wireless access technology operates. Low cost of access is an important aspect for popularizing a WLAN technology. Hence the design of WLAN should consider the parts of the frequency spectrum (*e.g.*, ISM band) for its operation which do not require an explicit licensing.

- **Tolerance to interference:** The proliferation of different wireless networking technologies both for civilian and military applications and the use of the microwave frequency spectrum for non-communication purposes (*e.g.*, microwave ovens) have led to a significant increase in the interference level across the radio spectrum. The WLAN design should account for this and take appropriate measures by way of selecting technologies and protocols to operate in the presence of interference.

- **Global usability:** The design of the WLAN, the choice of technology, and the selection of the operating frequency spectrum should take into account the prevailing spectrum restrictions in countries across the world. This ensures the acceptability of the technology across the world.

- **Security:** The inherent broadcast nature of wireless medium adds to the requirement of security features to be included in the design of WLAN technology.

- **Safety requirements:** The design of WLAN technology should follow the safety requirements that can be classified into the following: (i) interference to medical and other instrumentation devices and (ii) increased power level of transmitters that can lead to health hazards.

- **Quality of service requirements:** Quality of service (QoS) refers to the provisioning of designated levels of performance for multimedia traffic.

- **Compatibility with other technologies and applications:** The interoperability among the different LANs (wired or wireless).

WLANs can be broadly classified into two types, infrastructure networks and ad hoc LANs, based on the underlying architecture

Infrastructure networks contain special nodes called *access points* (APs), which are connected via existing networks. APs are special in the sense that they can interact with wireless nodes as well as with the existing wired network.

The other wireless nodes, also known as mobile stations (STAs), communicate via APs. The APs also act as bridges with other networks.

Ad hoc LANs do not need any fixed infrastructure. These networks can be set up on the fly at any place. Nodes communicate directly with each other or forward messages through other nodes that are directly accessible.

IEEE 802.11 is the most popular WLAN standard that defines the specification for the physical and MAC layers. The success of this standard can be understood from the fact that the revenue from the products based on this standard touched \$730 million in the second quarter of the year 2003.

The principles and mechanisms followed in this standard are explained later. In what follows, the basic components in a typical IEEE 802.11 WLAN

IEEE 802.11 STANDARD

WLANs. IEEE 802.11 is a prominent standard for WLANs, which is adopted by many vendors of WLAN products. A later version of this standard is the IEEE 802.11b , commercially known as *Wi-Fi* (wireless fidelity). The IEEE 802.11 standard, which deals with the physical and MAC layers in WLANs, was brought out in 1997.

As mentioned earlier, in contrast to other LAN standards, wireless standards need to have provisions to support mobility of nodes.

The IEEE802.11 working group had to examine connection management, link reliability management, and power management — none of which was a concern for other standards in IEEE 802. In addition, provision for security had to be introduced.

IEEE 802.11 standard, MTs can operate in two modes:

- (i) *infrastructure mode*, in which MTs can communicate with one or more APs which are connected to a WLAN, and
- (ii) *ad hoc mode*,

Physical Layer

IEEE 802.11 supports three options for the medium to be used at the physical level — one is based on infrared and the other two are based on radio transmission.

The physical layer is subdivided conceptually into two parts —

physical medium dependent sublayer (PMD) and
physical layer convergence protocol (PLCP).

PMD handles encoding, decoding, and modulation of signals and thus deals with the idiosyncrasies of the particular medium.

The PLCP abstracts the functionality that the physical layer has to offer to the MAC layer.

PLCP offers a service access point (SAP) that is independent of the transmission technology, and a clear channel assessment (CCA) carrier sense signal to the MAC layer.

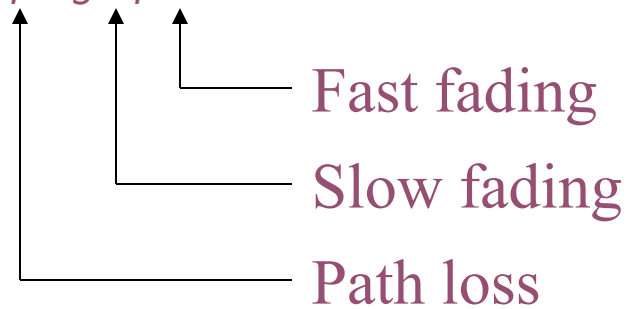
Land Propagation

- The received signal power:

$$P_r = \frac{G_t G_r P_t}{L}$$

where G_r is the receiver antenna gain,

L is the propagation loss in the channel, i.e.,

$$L = L_P L_S L_F$$


Fast fading

Slow fading

Path loss

Radio Channel Features*

- ***Asymmetrical links***: connectivity from node a to node b might differ significantly from b to a
- ***Non-isotropical connectivity***: connectivity need not be same in all directions (at same distance from source)
- ***Non-monotonic distance decay***: nodes geographically far away from source may get better connectivity than nodes that are geographically closer

Parameters

- ***Transmission gain control***: most mobile nodes low power radios have some form TX gain control
- ***Antenna height***: relative distance of antenna wrt reference ground
- ***Radio frequency and modulation type***
- ***Packet size***: # bits per packet can affect likelihood of receiving the packet with no errors
- ***Data rate***: # packets transmitted per second
- ***Environment type***: e.g., indoors or outdoors, with or w/o LOS, different levels of physical interference (furniture, walls, trees, etc.), and different materials (sand, grass, concrete, etc.)